



وفاقی اردو یونیورسٹی
برائے فنون، سائنس و ٹیکنالوجی، اسلام آباد

NAME: Uzma Essa

SEAT NO.: 23122038

SEMESTER: 6TH

SECTION: B

SUBJECT: Artificial Intelligence

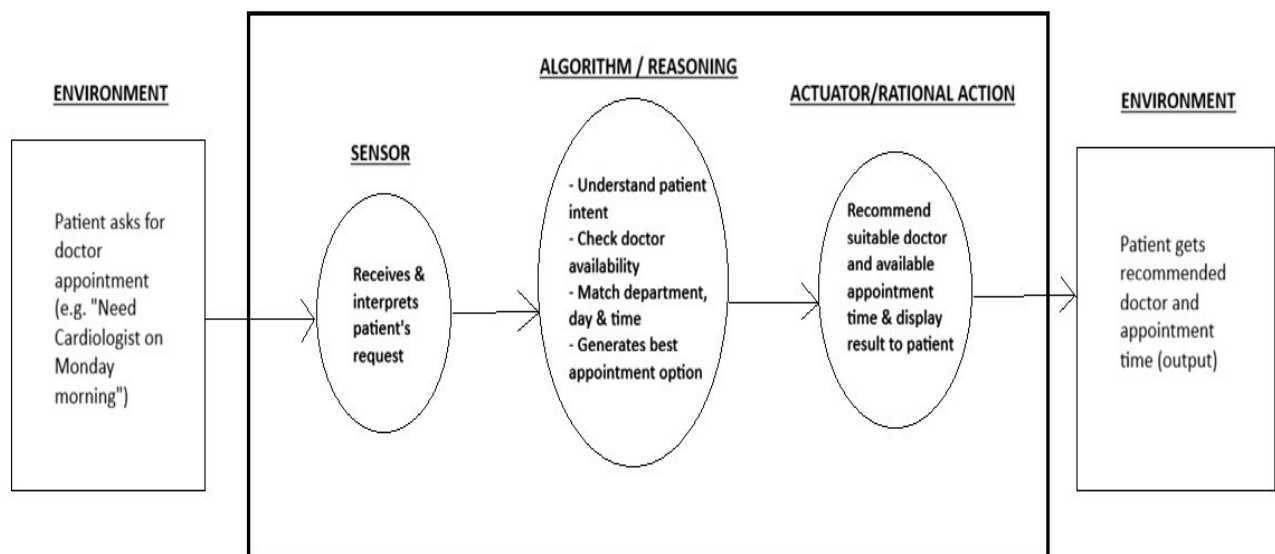
TEACHER: Ma'am Uzma

AI / INTELLIGENT AGENT ASSIGNMENT

AGENT NAME: Smart Doctor Appointment Agent

PROBLEM FORMULATION: Recommend a suitable doctor and available appointment time to patients based on their required department, preferred day, and preferred time.

GRAPHICAL REPRESENTATION:



AGENT FORMULATION:

Environment: Hospital appointment system, doctors, patients, departments and available time slots.

Sensors: Patient request, department, preferred doctor, preferred day/time and doctor availability.

Knowledge Base: Doctor names, departments, working days, available time slots, and appointment availability.

Algorithm/reasoning: Understands the request, use knowledge base (database schema) to check doctor availability and matches the best options.

Actuator / Rational Action: Recommend suitable doctor, and appointment time and display result to the patient.

Example:

Doctor	Department	Day	Time	Availability
Dr. Ahmed	Cardiology	Monday	Morning	Available
Dr. Sara	Dermatology	Tuesday	Evening	Available
Dr. Ali	General	Monday	Evening	Available
Dr. Hina	Pediatrics	Thursday	Morning	Available

ENVIRONMENT CLASSIFICATION:

Partially Observable: The agent can see patient information and available appointment slots, but it may not know unexpected changes, such as a doctor suddenly becoming unavailable.

Deterministic: With the same patient requirements and availability data, the agent will make the same decision using the same rules.

Sequential: One booking can affect later decisions because once a slot is booked, it is no longer available.

Dynamic: Doctor availability and appointment slots can change while the agent is operating.

Discrete: Doctors, departments, days, and appointment slots are separate choices, such as Cardiology, Monday, Morning.

Known: The agent knows the available doctors, departments, appointment slots, and rules stored in the database.

Multi-agent: Patients, doctors, hospital staff, and the appointment agent interact within the environment.

PEAS EVALUATION:

Performance Measure: Correct matching, appointment availability, reduced waiting, quick response, and patient satisfaction.

Environment: Hospital, doctors, patients, departments, appointment database and time slots.

Actuator: Display recommendation, offer another slot, confirm or book the appointment.

Sensor: Patient request, department, preferred day/time and doctor availability.